

ITI AI-Pro

Natural Language Processing



CONTENTS

- Large Language Models
- Transfer Learning
- Cutting-Edge Fine-Tuning
- Productization
- Hugging Face
- Main Resources
- RAG
- Open Space

Part I

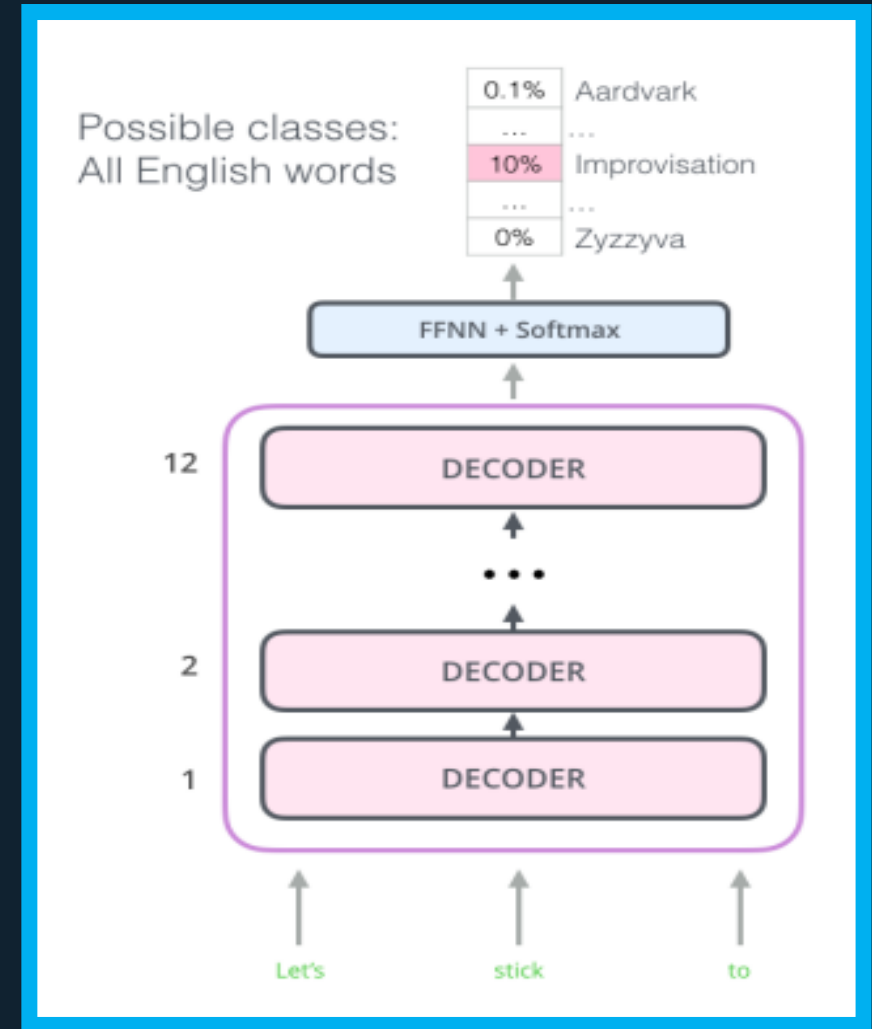
Large Language Models (LLMs)



GPT

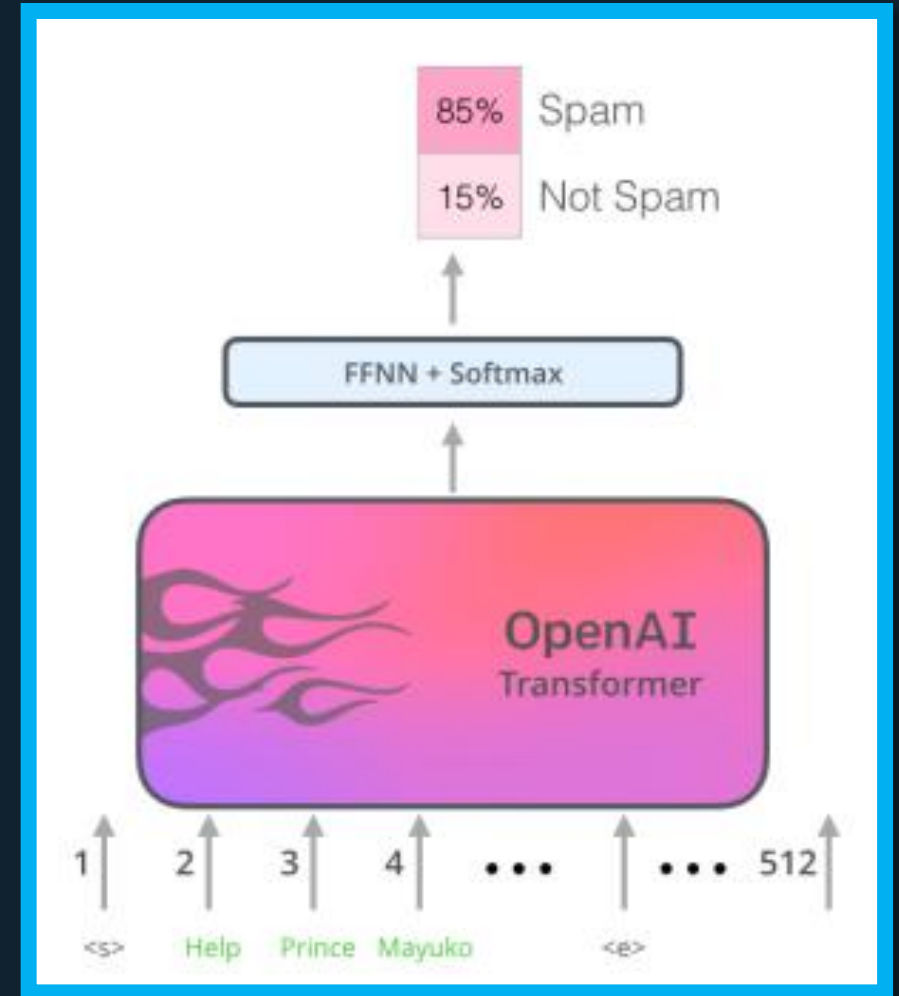
- It uses a stack of **12 decoder** unit.
- **No encoder** is used.
- The objective is to **predict the next word**

[More](#)



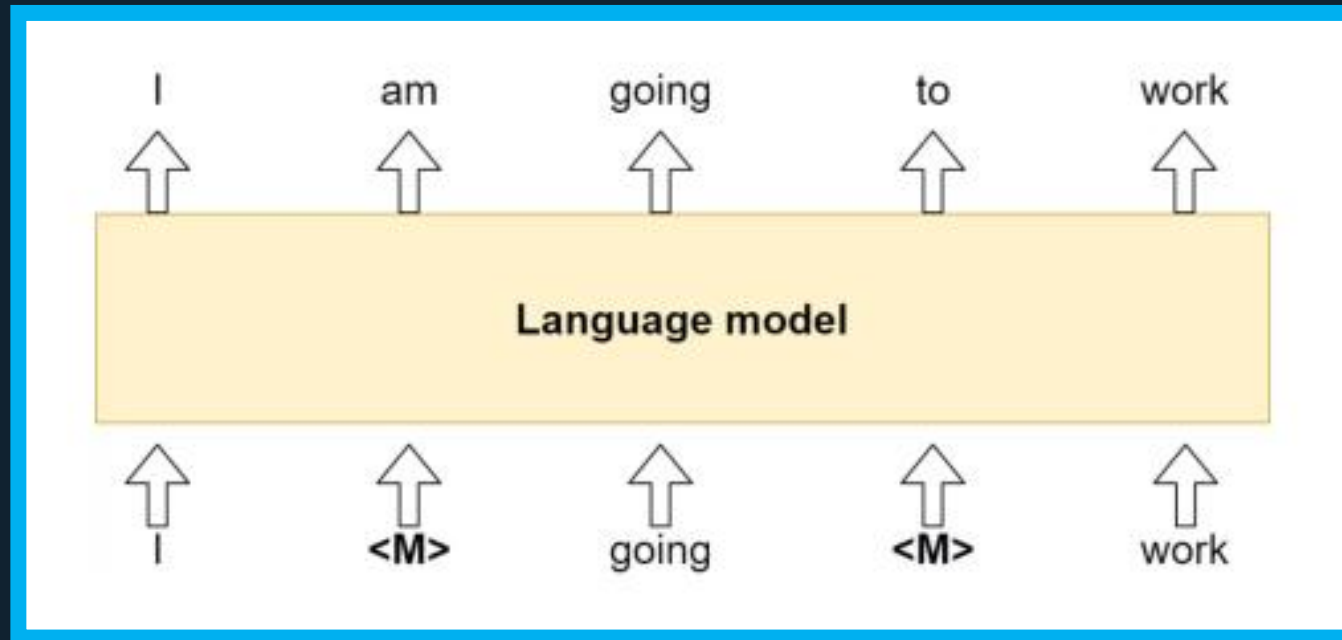
GPT

- Decoder has a **drawback**, it can't attend to **future tokens**.



BERT

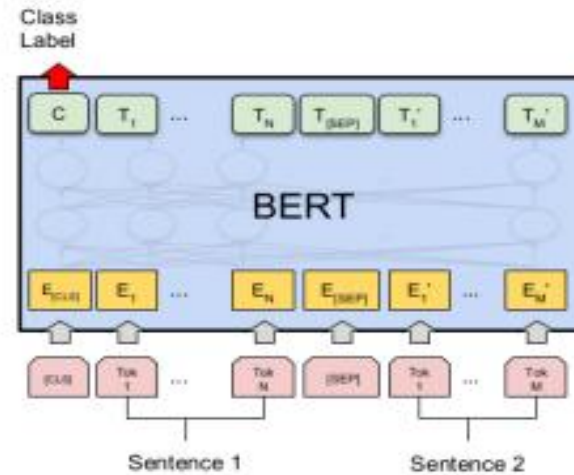
- **Encoders** instead of Decoders
- Trained on masked language modeling task
- BERT objective is to **predict masked token**



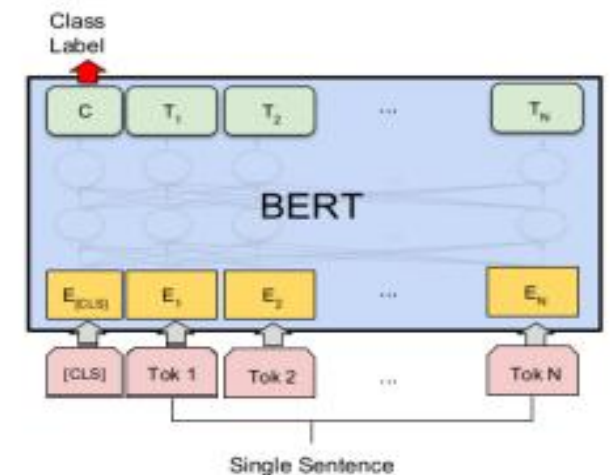
[More](#)

BERT

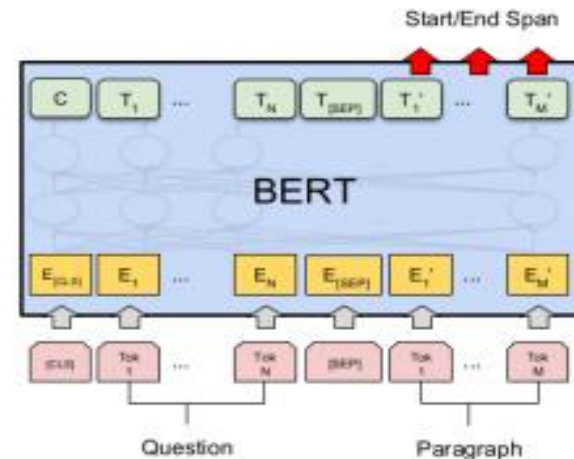
- BERT can be used for almost **all NLP task**



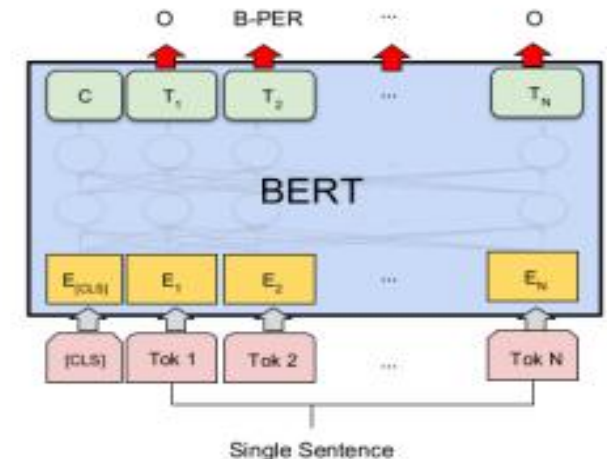
(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



(b) Single Sentence Classification Tasks:
SST-2, CoLA



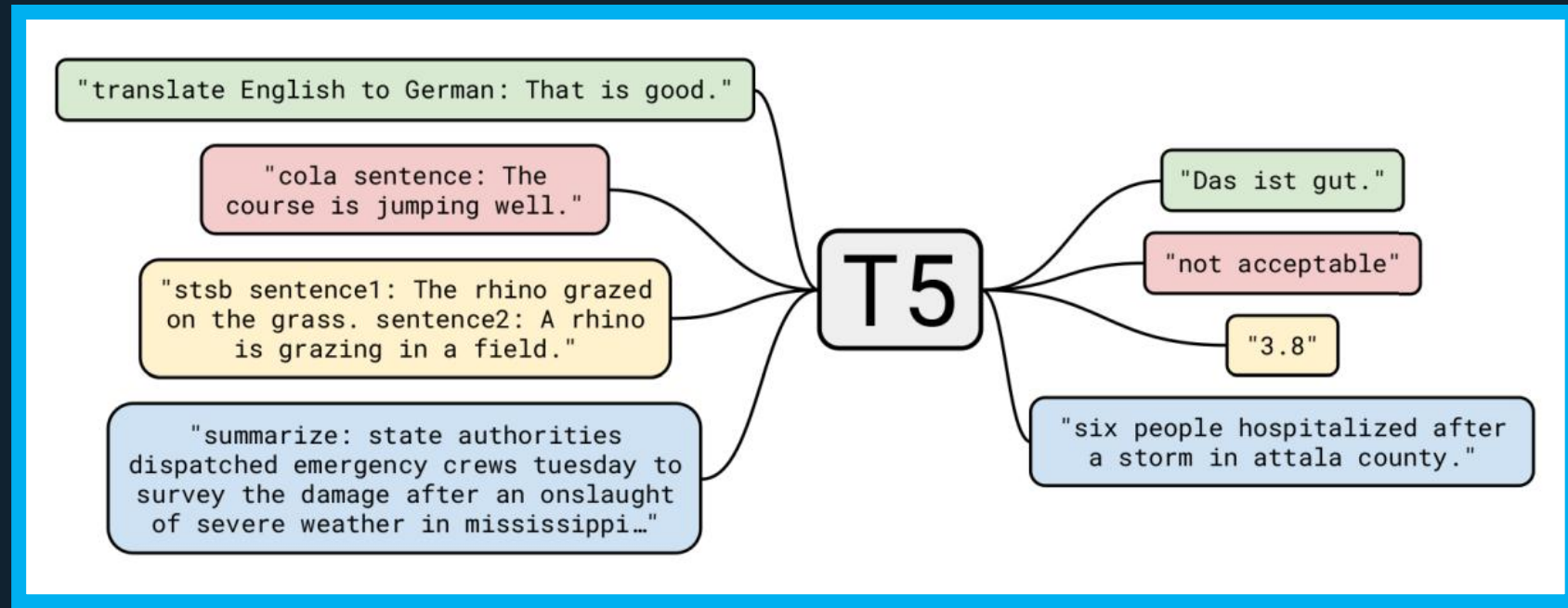
(c) Question Answering Tasks:
SQuAD v1.1



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

T5

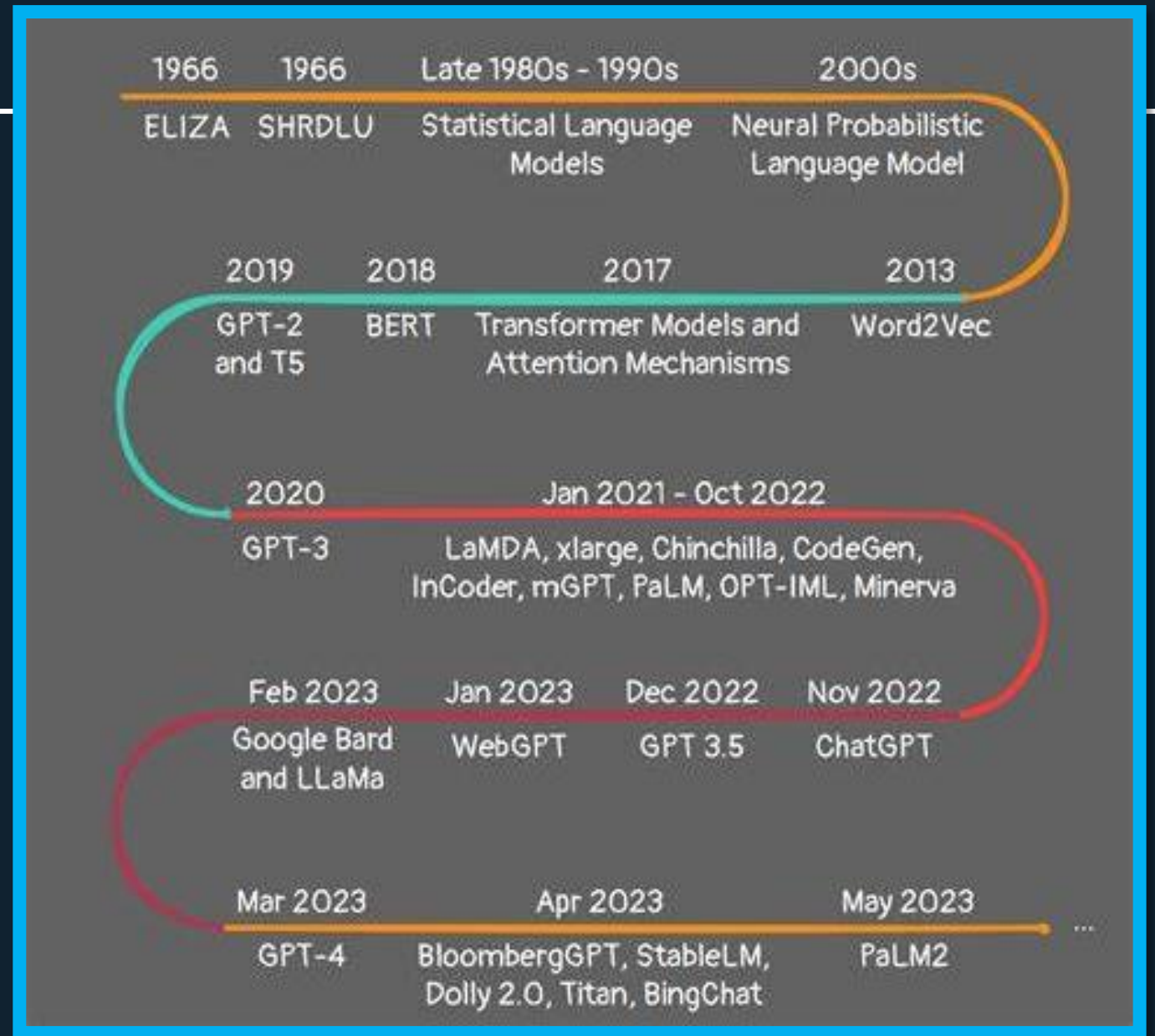
- Google 2020 : **encoder-decoder** architecture
- Trained on the **Colossal Clean Crawled Corpus (C4)**
huge corpus covering **differing** kind of **tasks**



[More](#)

LLMS TIMELINE

- The most critical point
ChatGPT Nov 2022



OTHER LLMS

- LLaMA
- Falcon
- Bard
- Gemini
- Mistral
- Mixtral
- ... etc.

[More](#)



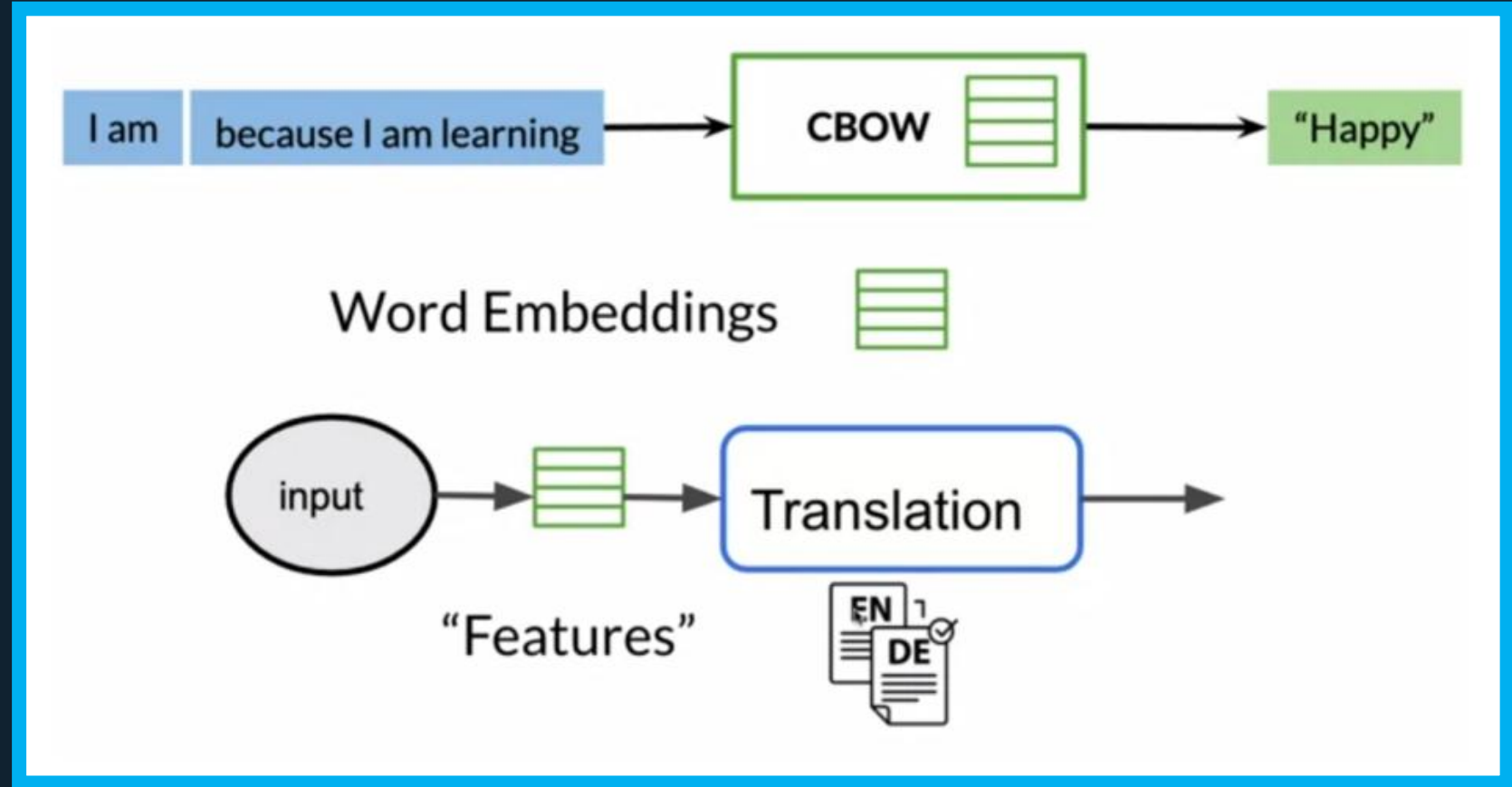
Part II

Transfer Learning



TRANSFER LEARNING

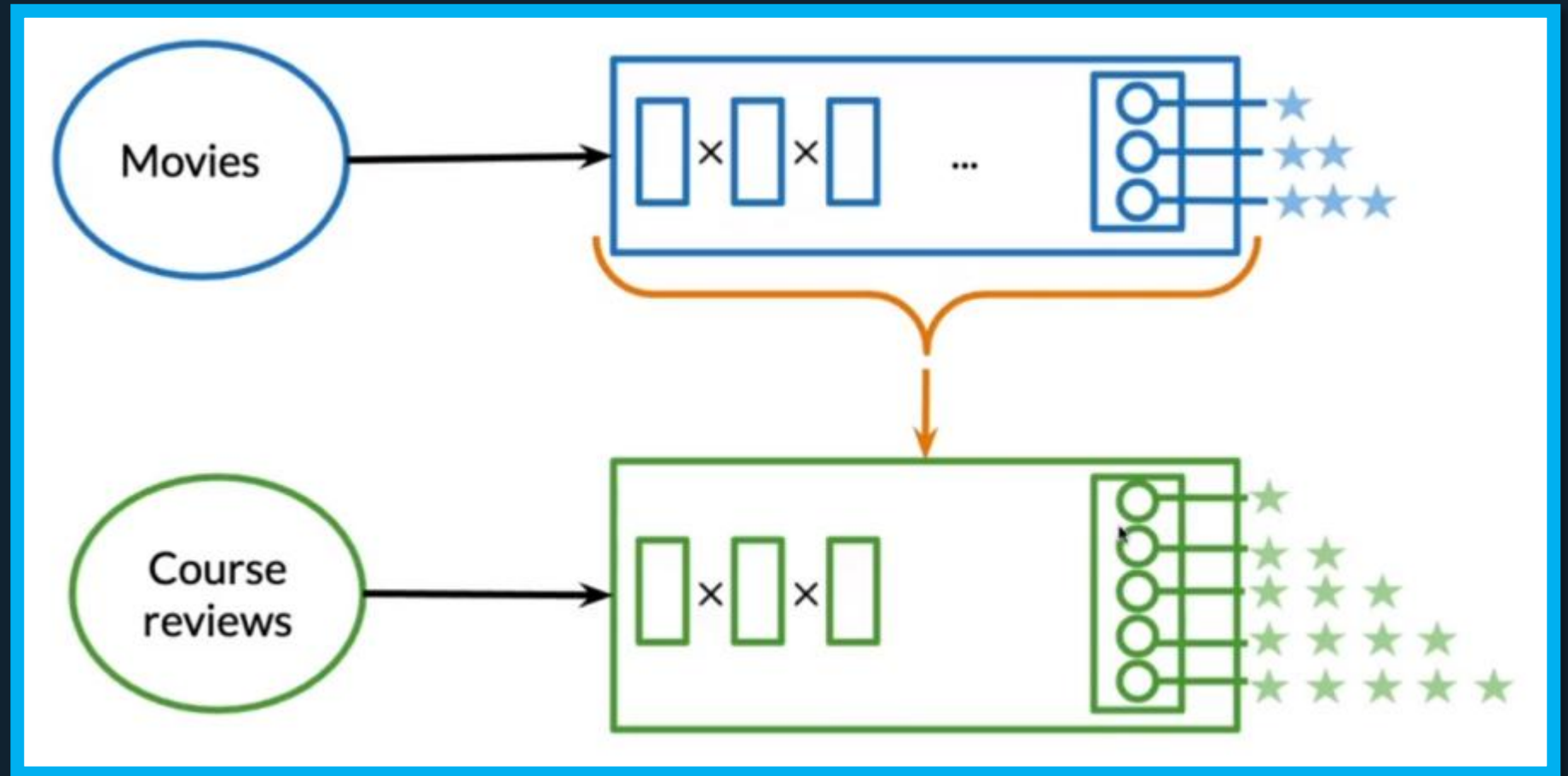
- **Feature – Based Transfer Learning**



[More](#)

TRANSFER LEARNING

- **Fine – Tuning** Transfer Learning

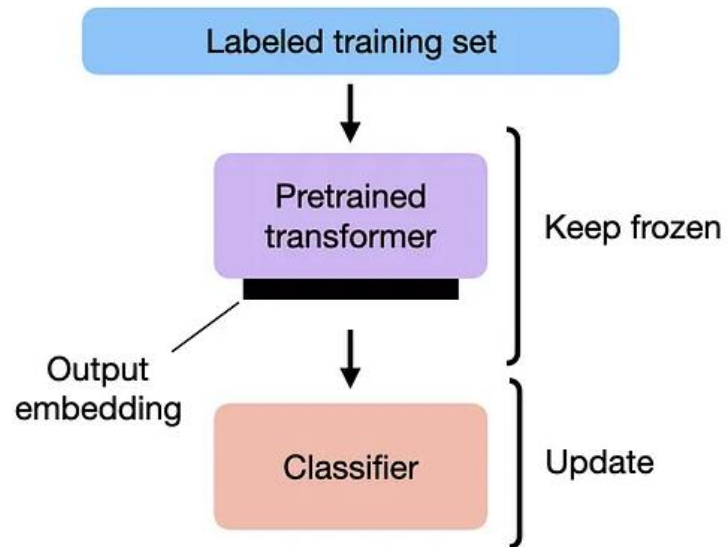


[More](#)

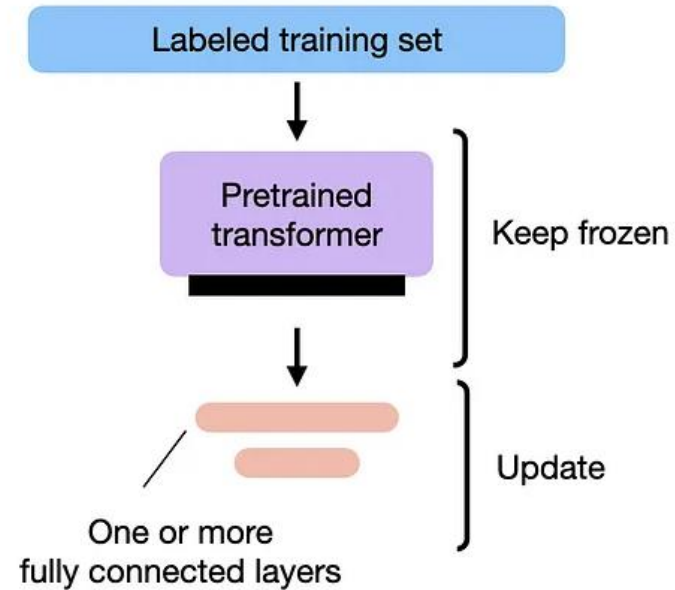
TRANSFER LEARNING

- Old School

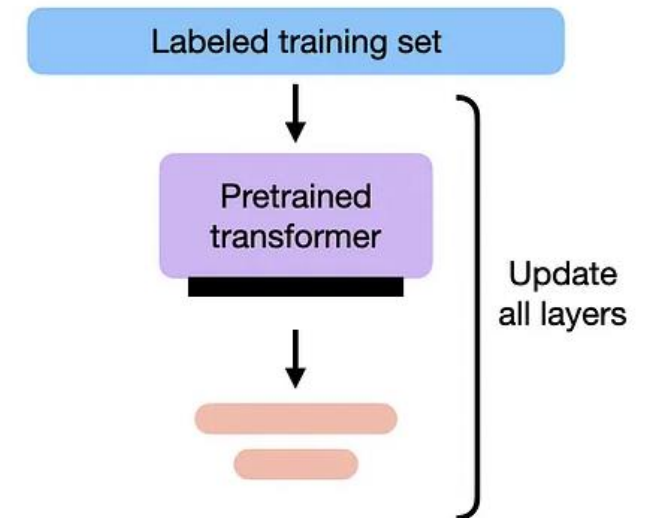
1) FEATURE-BASED APPROACH



2) FINETUNING I



3) FINETUNING II





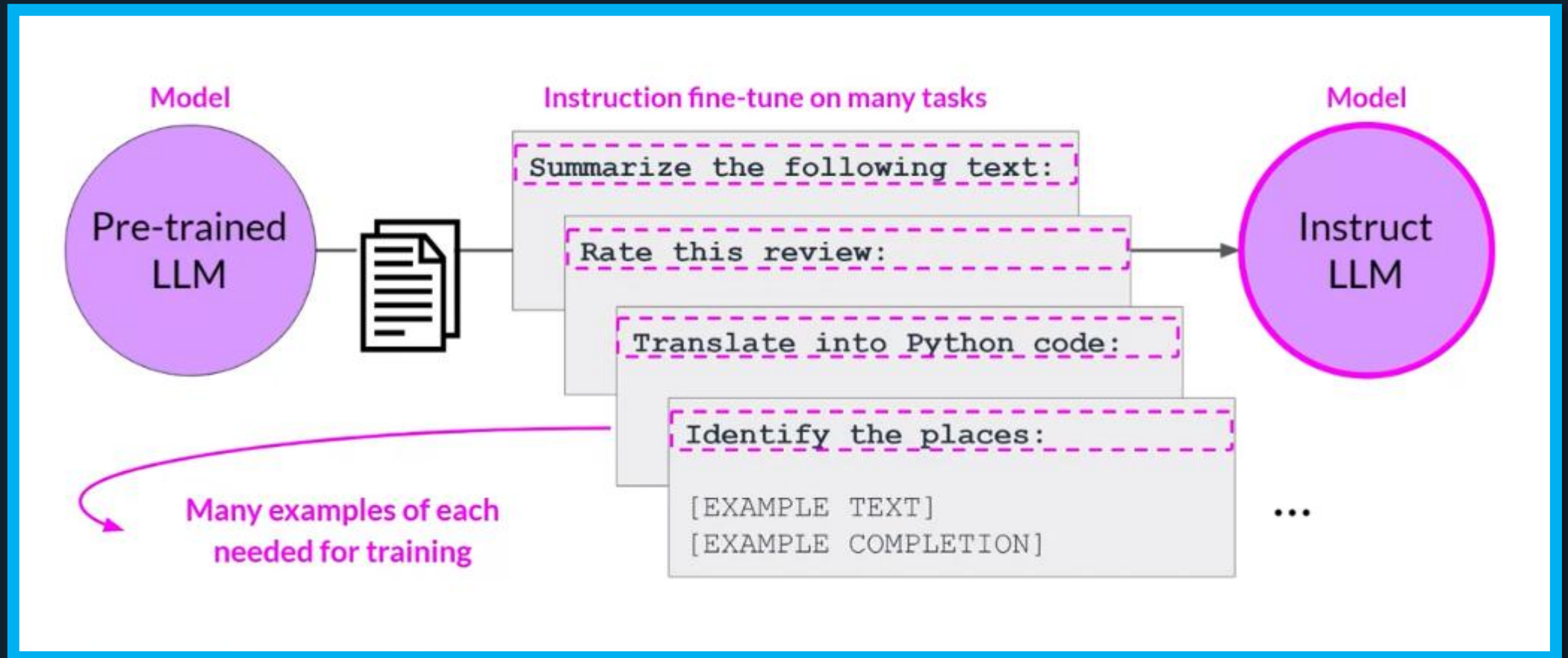
Part III

Cutting-Edge Fine-Tuning



CUTTING-EDGE FINE-TUNING

- Instruction Tuning



[More](#)

CUTTING-EDGE FINE-TUNING

- Instruction Tuning

```
{  
  "instruction": "Write a limerick about a  
                 pelican.",  
  "input": "",  
  "output": "There once was a pelican so fine,  
             \nHis beak was as colorful as  
             sunshine,\nHe would fish all day,\nIn  
a very unique way,\nThis pelican was  
truly divine!\n\n\n",  
},
```

Input is optional

```
{  
  "instruction": "Identify the odd one out from  
                 the group.",  
  "input": "Carrot, Apple, Banana, Grape",  
  "output": "Carrot\n\n",  
},
```

CUTTING-EDGE FINE-TUNING

- Chain-of-thought (CoT) Fine-Tuning

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

[More](#)

CUTTING-EDGE FINE-TUNING

- **Task Specific Fine-Tuning**

todavía está en casa → Are you still at home?

- **Instruction Tuning**

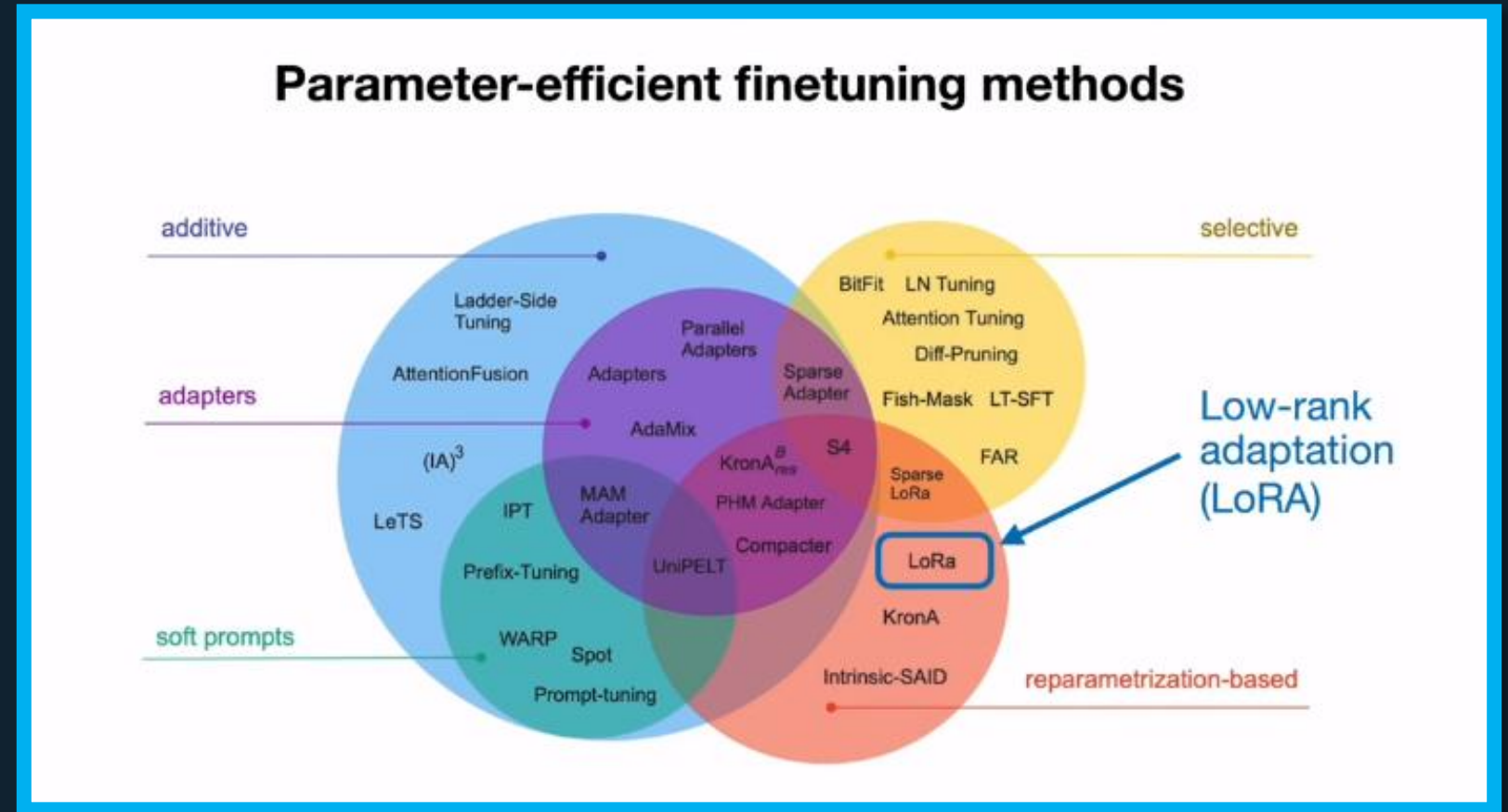
Translate this sentence into English “todavía está en casa”

→ Are you still at home?

[More](#)

CUTTING-EDGE FINE-TUNING

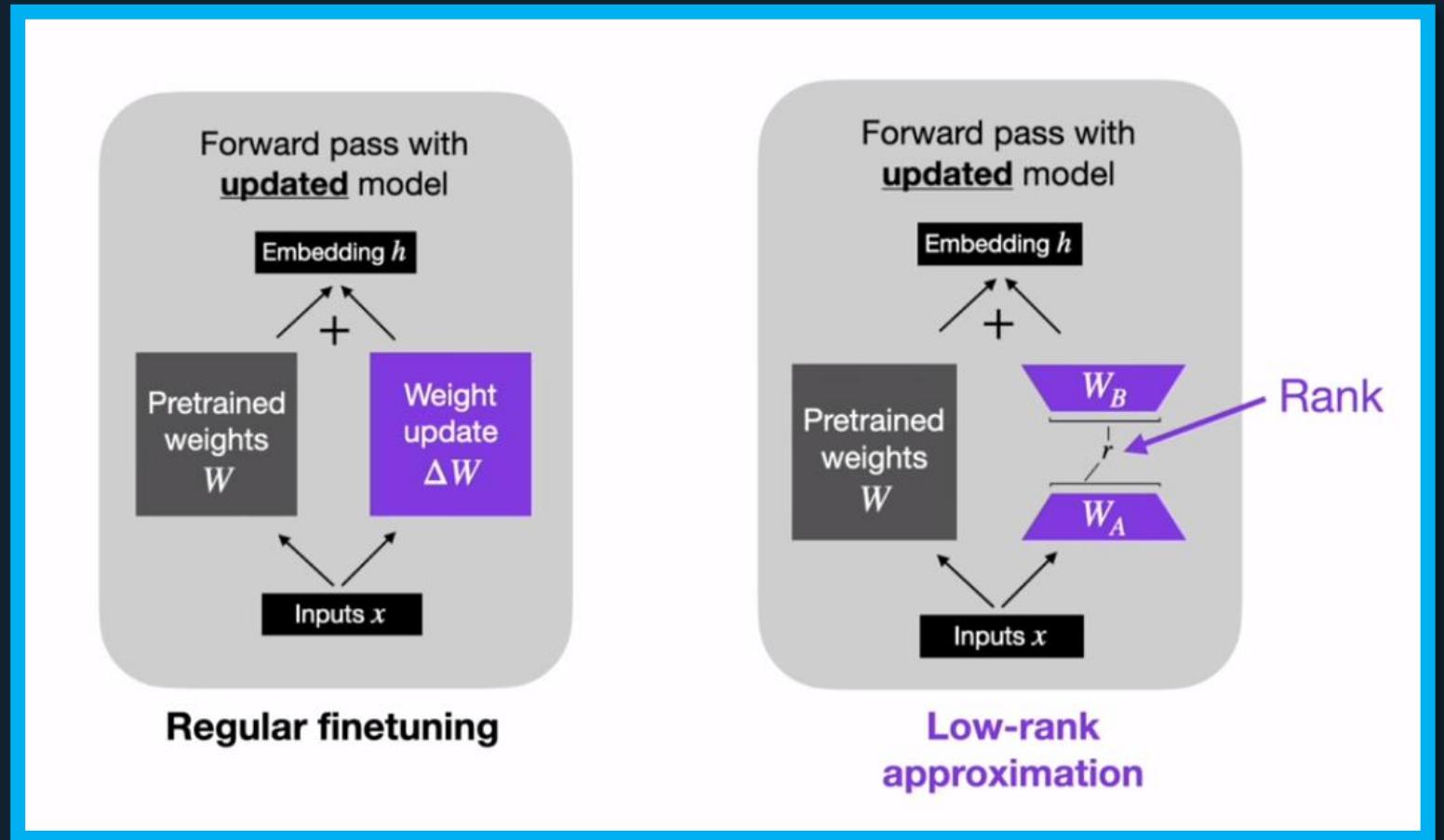
- **LoRA: Low-Rank Adaptation**



[More](#)

CUTTING-EDGE FINE-TUNING

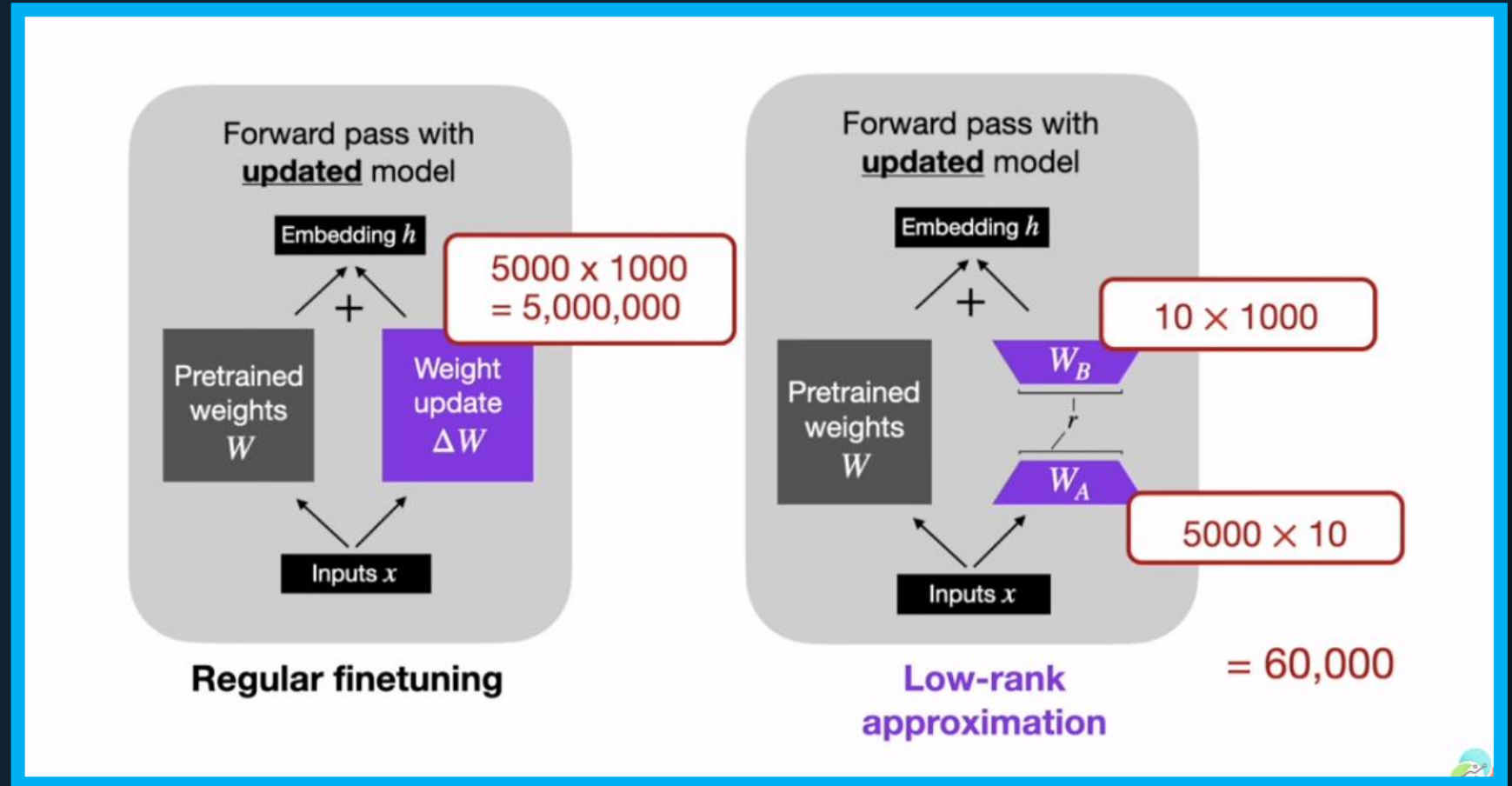
- **LoRA: Low-Rank Adaptation**



[More](#)

CUTTING-EDGE FINE-TUNING

- LoRA: Low-Rank Adaptation



[More](#)

CUTTING-EDGE FINE-TUNING

- **QLoRA: Quantized Low-Rank Adaptation**

LoRA + Quantization

4-bit weights

(compared to **8-bits** in the case of LoRA)

[More](#)



Part IV

Productization



TRANSFORMERS LIMITATIONS

Large Size

+

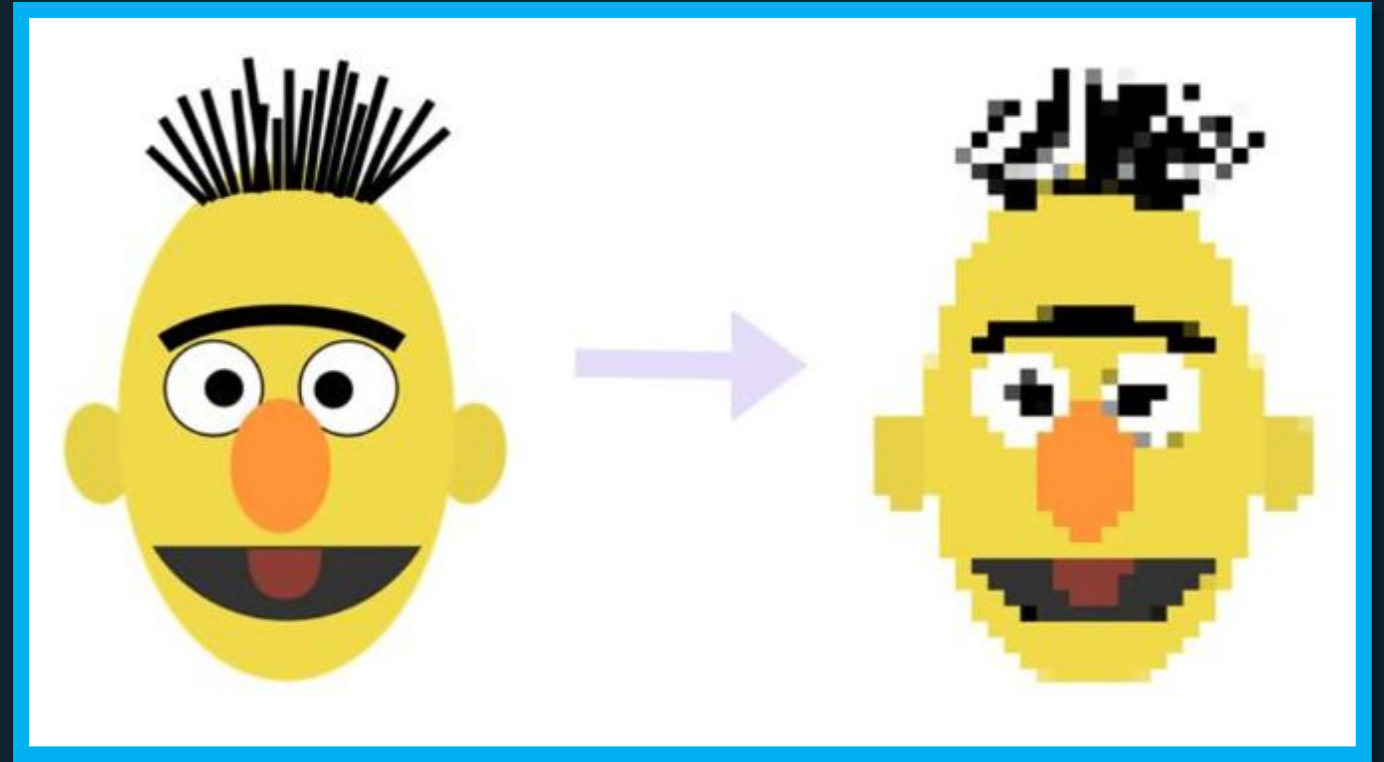
Computational Power

+

GPU for Inference Time

PRODUCTIZATION

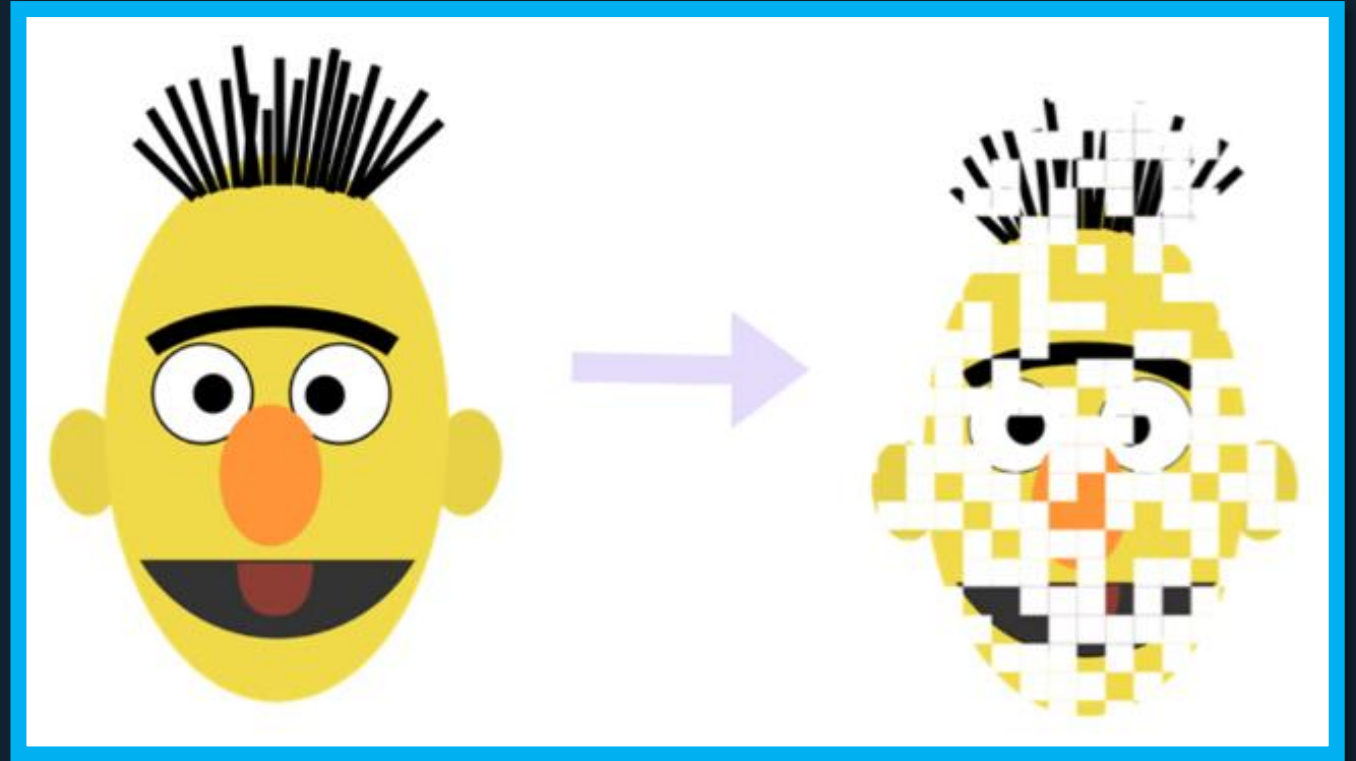
- **Quantization** reducing the precision of the weights to **8-bits** reduce **memory usage** and the **accuracy** of the model



[More](#)

PRODUCTIZATION

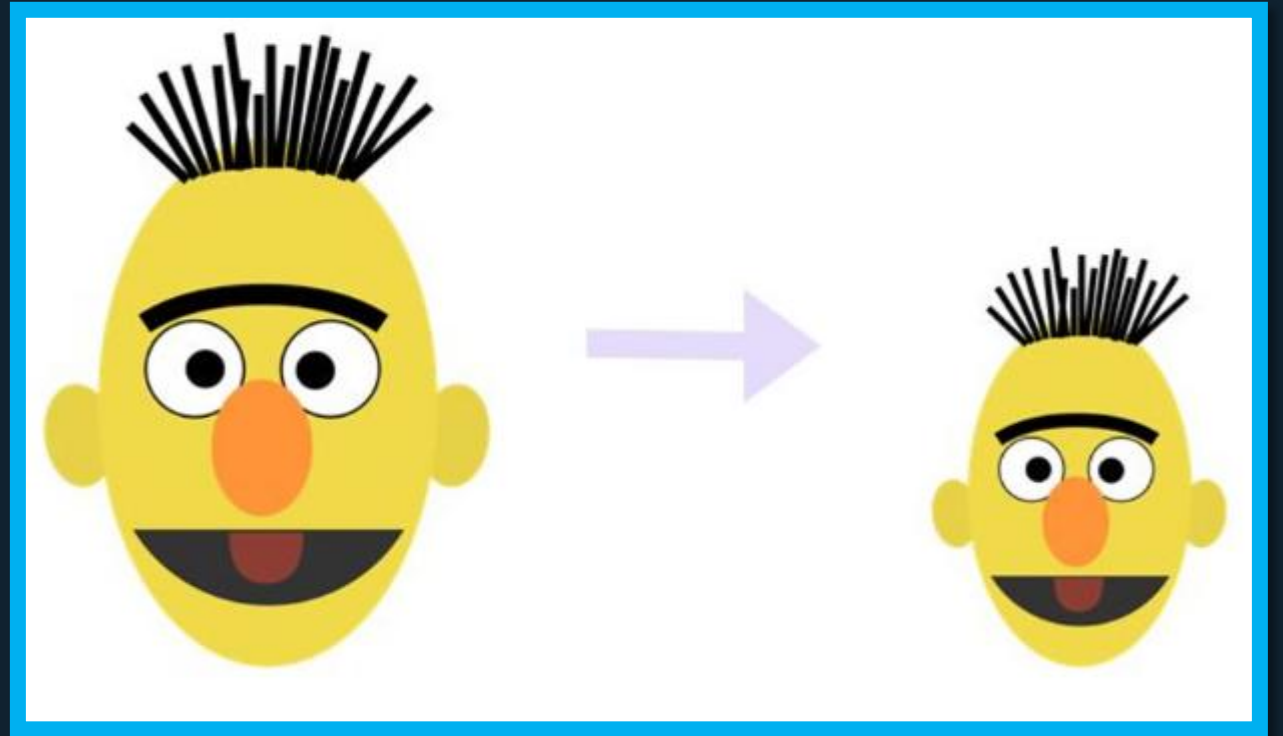
- **Pruning** remove some **weights** that is very **insignificant**
 - when it is **close to 0** it's **not important**
 - removes **neurons** or even a **complete weight matrix**



[More](#)

PRODUCTIZATION

- **Knowledge Distillation** after training a big and slow model (teacher), a **smaller model** (student) is trained to **mimic** the teacher's behavior
- The student trains to **copy the knowledge** of the teacher.



[More](#)



Hugging Face



NBs I

Hugging Face
Notebooks



NB II

Sentence
Transformers



NB III
ONNX



Open LLMs



Arena AI

(Leaderboard & Chat)



Part VII

RAG



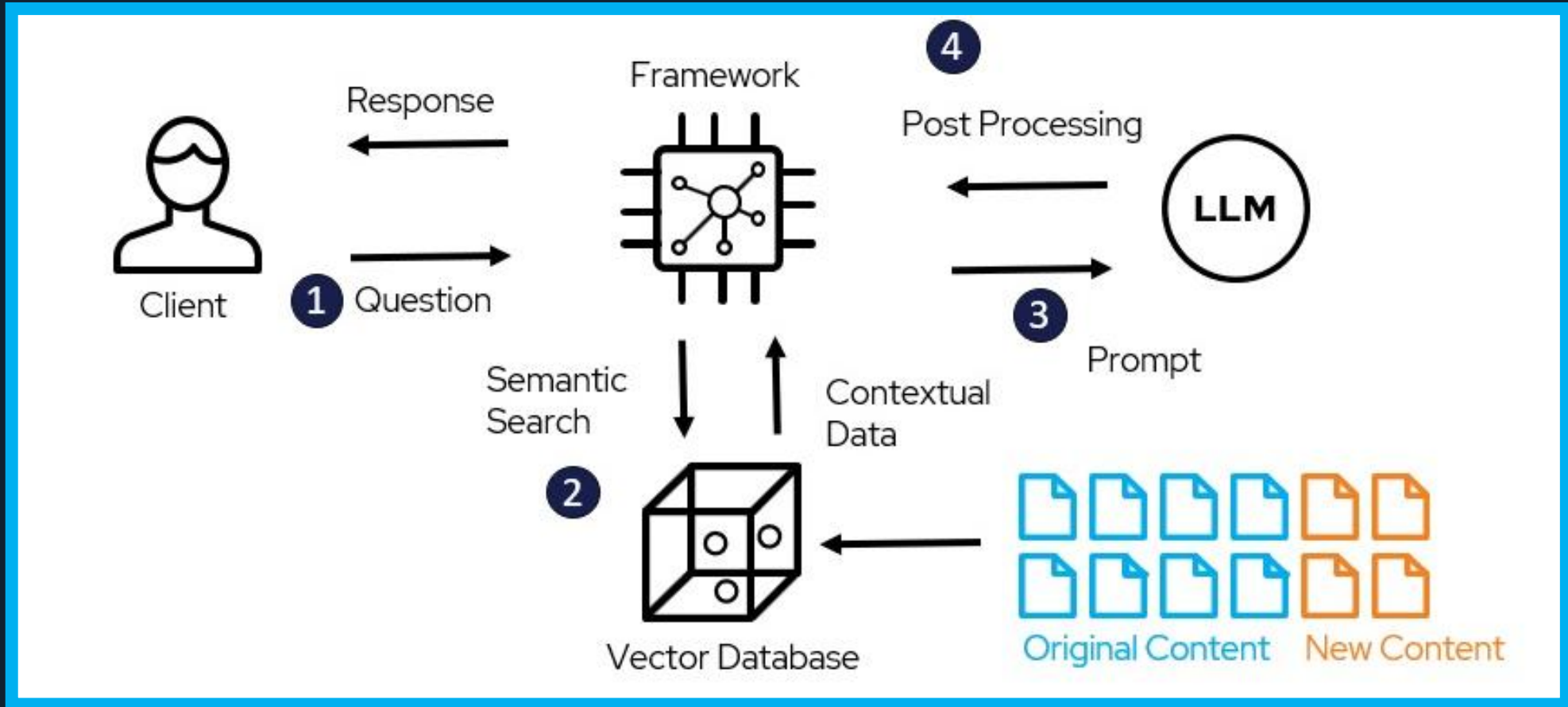
RAG

- Retrieval Augmented Generation
 - Knowledge Mining



RAG

- Utilize LLM capabilities without fine-tuning

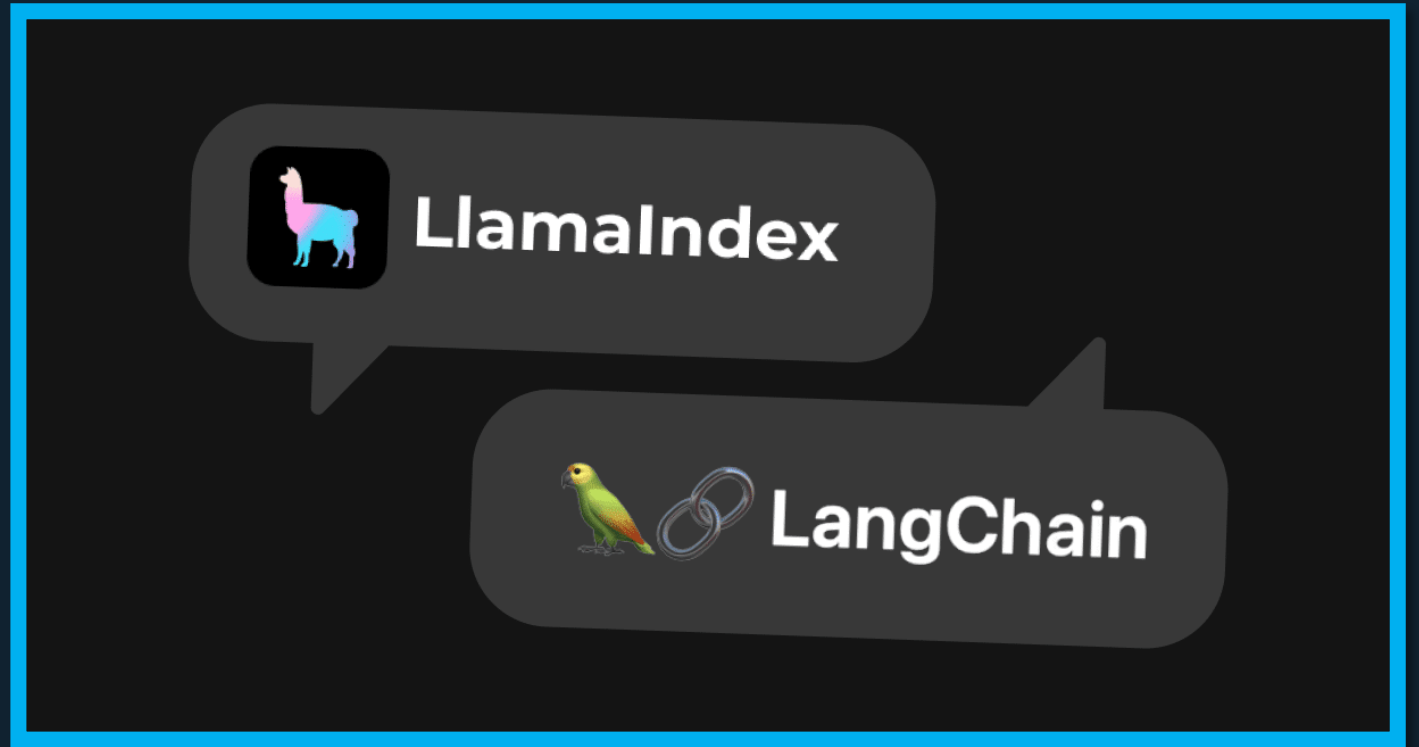


RAG

- Frameworks:

[LangChain](#)

[Llama-Index](#)





Open Space



THANKS
A LOT